



**Intelligent
Embedded Systems**

Cats AI — current status

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- Identified a suitable crash-video dataset with normal and accident clips.
- Built a Qwen-VL video querying pipeline for structured accident analysis.
- Designed JSON prompts and validation schema for consistent model outputs.
- Ran first evaluation experiments measuring valid JSON rate and crash-detection accuracy.
- Prepared the basic SFT data format and LoRA training pipeline.

- Task: classify whether a dashcam video contains a visible accident.
- Input: sampled video frames from normal and crash videos.
- Output: structured JSON with accident prediction.
- Metrics:
 - Valid JSON outputs: **100%**
 - Detection accuracy: **92.7%**
- Result: the pipeline works end-to-end and can be used for larger-scale experiments.

1 Dataset with analysis labels

Find a dataset that already contains accident explanations, risk factors, or scene-level reasoning labels.

2 Knowledge distillation

Use a stronger teacher model to generate structured assistant JSONs, then fine-tune the local model on these outputs.

3 Hybrid approach

Use datasets with partial analysis labels and let the teacher model enrich or complete them.

- For now, we choose **knowledge distillation**.
- Reason: no suitable dataset with the required analysis-style labels has been found yet.
- The teacher model will generate structured JSON outputs:
 - accident presence
 - confidence
 - scene description
 - main objects
 - risk factors
 - reasoning
 - uncertainty

- Generate assistant JSONs from a stronger teacher model.
- Validate generated JSONs against the schema.
- Filter low-quality or incorrect outputs using the existing crash/normal labels.
- Build the distilled SFT dataset.
- Run a small LoRA SFT experiment and compare before vs. after performance.